Learn with AWS
From classroom to cloud: Join AWS Builder Center to access learning paths with integrated AI tools to accelerate your learning.

Joi

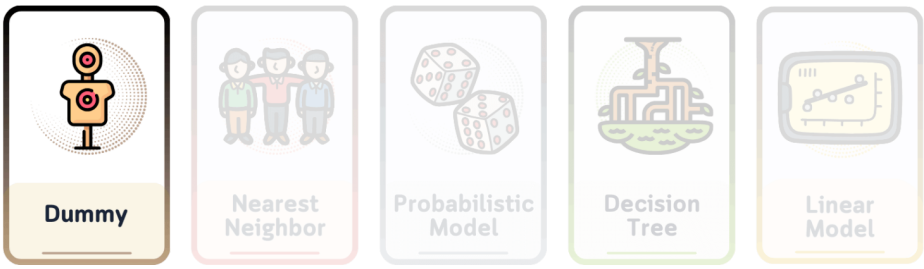
MACHINE LEARNING

Dummy Classifier Explaine Visual Guide with Code Ex for Beginners

Setting the Bar in Machine Learning with Simple Base

Samy Baladram
Aug 14, 2024 7 min read

CLASSIFICATION ALGORITHM

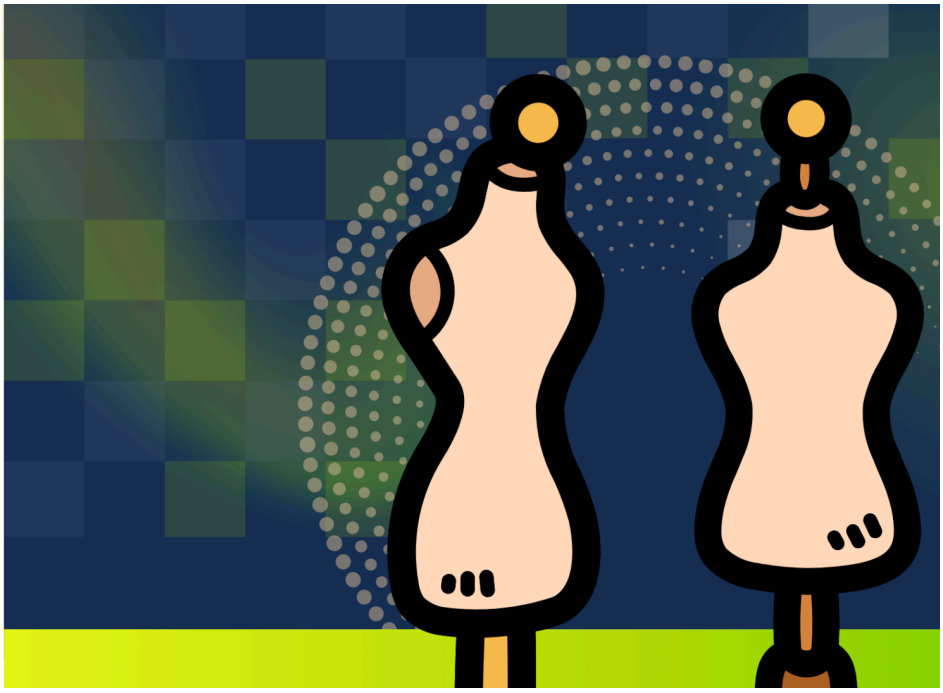


⛳ More CLASSIFICATION ALGORITHM, explained: ▶️ [Dumm
(https://towardsdatascience.com/dummy-classifier-expla
guide-with-code-examples-for-beginners-009ff95fc86e)
Neighbor Classifier](https://towardsdatascience.com/k-

Learn with AWS
From classroom to cloud: Join AWS Builder Center to learning paths with integrated AI tools to accelerate learning.

guide-with-code-examples-for-beginners-aec39771ddd6) · [Gaussian Naive Bayes](https://towardsdatascience.com/gaussian-naive-bayes-explained-a-visual-guide-with-code-examples-for-beginners-04949cef383c) · [Decision Tree Classifier](https://towardsdatascience.com/decision-tree-classifier-explained-a-visual-guide-with-code-examples-for-beginners-7c863f06a71e) · [Logistic Regression](https://towardsdatascience.com/logistic-regression-explained-a-visual-guide-with-code-examples-for-beginr
[Support Vector Classifier](https://towardsdatascience
vector-classifier-explained-a-visual-guide-with-mini-2
62e831e7b9e9) · [Multilayer Perceptron]
(https://towardsdatascience.com/multilayer-perceptron-
guide-with-mini-2d-dataset-0ae8100c5d1c)

Have you ever wondered how data scientists me
performance of their machine learning models?
classifier – a simple yet powerful tool in the wo
science. Think of it as the baseline player in a ga
minimum standard that other, more sophisticat
to beat.



All visuals: Author-created using Canva Pro. Optimized for mobile; may appear oversized on desktop.

Definition

A dummy classifier is a simple machine learning model that makes predictions using basic rules, without actually learning from the input data. It serves as a baseline for c performance of more complex models. The dum helps us understand if our sophisticated model learning useful patterns or just guessing.

Dummy Classifier is one of the basic key algorithms in machi

📊 Dataset & Libraries

Throughout this article, we'll use this simple art (inspired by [1]) as an example. This dataset pre person will play golf based on weather conditior features like outlook, temperature, humidity, an target variable being whether to play golf or not.

Training Set

0	85	85	0	No
1	80	90	1	No
2	83	78	0	Yes
3	70	96	0	Yes
4	68	80	0	Yes
5	65	70	1	No
6	64	65	1	Yes
7	72	95	0	No
8	69	70	0	Yes
9	75	10	0	Yes
10	75	70	1	Yes
11	72	90	1	Yes
12	81	75	0	Yes
13	71	80	1	No

Test Set

14	81	88	1	No
15	74	92	0	Yes
16	76	85	0	Yes
17	78	75	1	No
18	82	92	0	No
19	67	9		
20	85	8		
21	73	8		
22	88	6		
23	77	7		
24	79	6		
25	80	9		
26	66	7		
27	84	7		

LATEST
EDITOR'S PICKS
DEEP DIVES
NEWSLETTER
WRITE FOR TDS
Submit an Article

Columns: 'Outlook', 'Temperature', 'Humidity', 'Wind' and 'Play' (

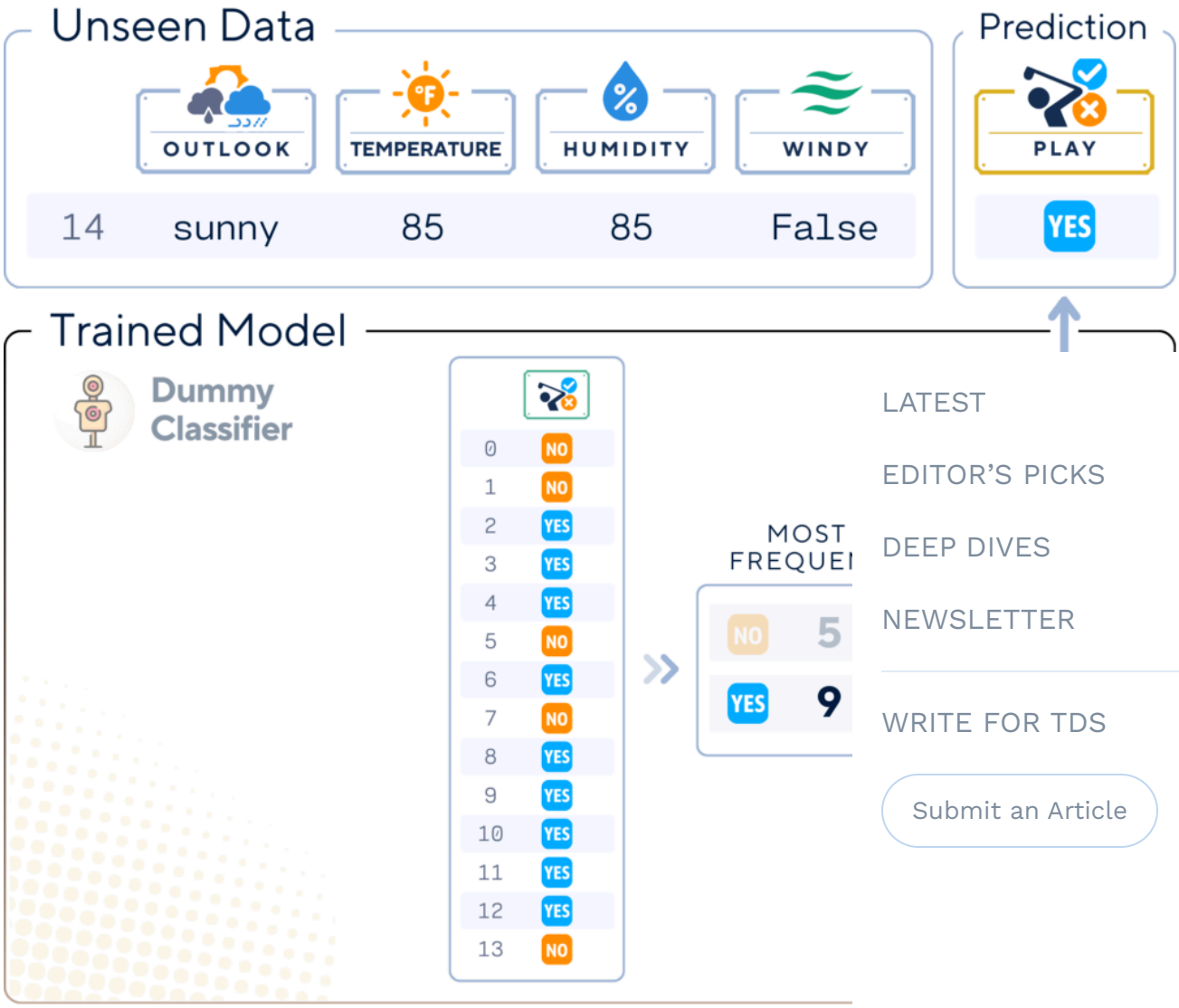
```
# Import libraries
from sklearn.model_selection import train_test_split
import pandas as pd

# Make a dataset
dataset_dict = {
    'Outlook': ['sunny', 'sunny', 'overcast', 'rain'
    'Temperature': [85.0, 80.0, 83.0, 70.0, 68.0, 65
    'Humidity': [85.0, 90.0, 78.0, 96.0, 80.0, 70.0,
    'Wind': [False, True, False, False, False, True,
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No',
}
df = pd.DataFrame(dataset_dict)

# One-hot Encode 'Outlook' Column
df = pd.get_dummies(df, columns=['Outlook'],  prefix

# Convert 'Windy' (bool) and 'Play' (binary) Columns
df['Wind'] = df['Wind'].astype(int)
```

```
df['Play'] = (df['Play'] == 'Yes').astype(int)

# Set feature matrix X and target vector y
X, y = df.drop(columns='Play'), df['Play']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, train_size=0.5
```

Main Mechanism

The dummy classifier operates on simple strate
predictions. These strategies don't involve any a
from the data. Instead, they use basic rules like

1. Always predicting the most frequent class
2. Randomly predicting a class based on the tr
   distribution
3. Always predicting a specific class



For our golf dataset, a dummy classifier might always predict "Yes" for playing g
outcome in the training data.

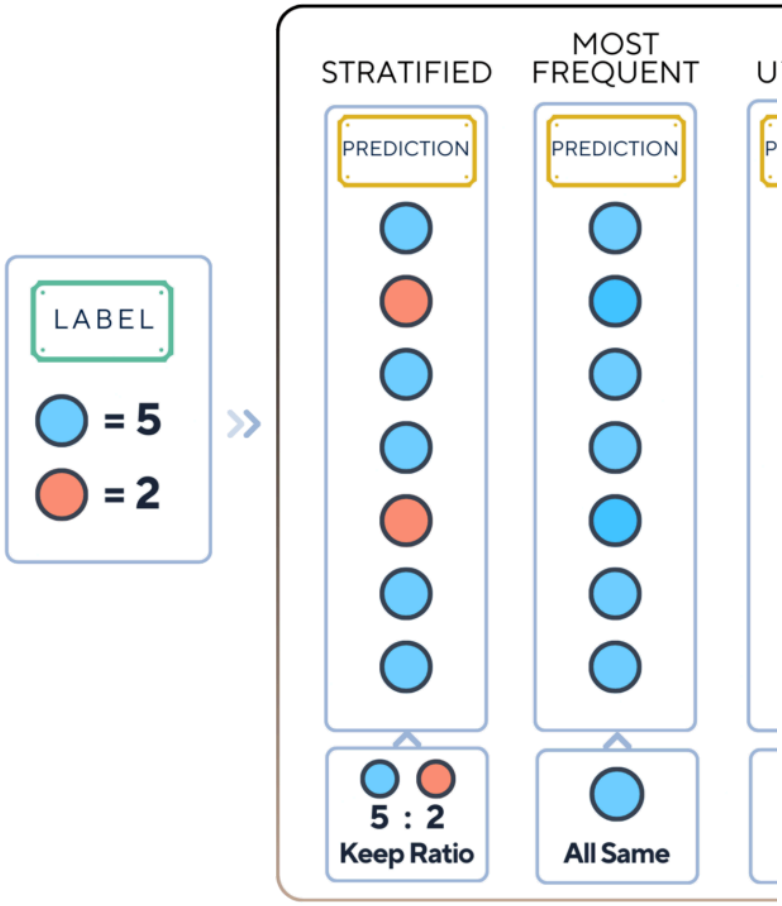
Training Steps

The "training" process for a dummy classifier is
doesn't involve the usual learning algorithms. He
outline:

1. Select Strategy

Choose one of the following strategies:

- **Stratified**: Makes random guesses based on the original class distribution.
- **Most Frequent**: Always picks the most common class.
- **Uniform**: Randomly picks any class.

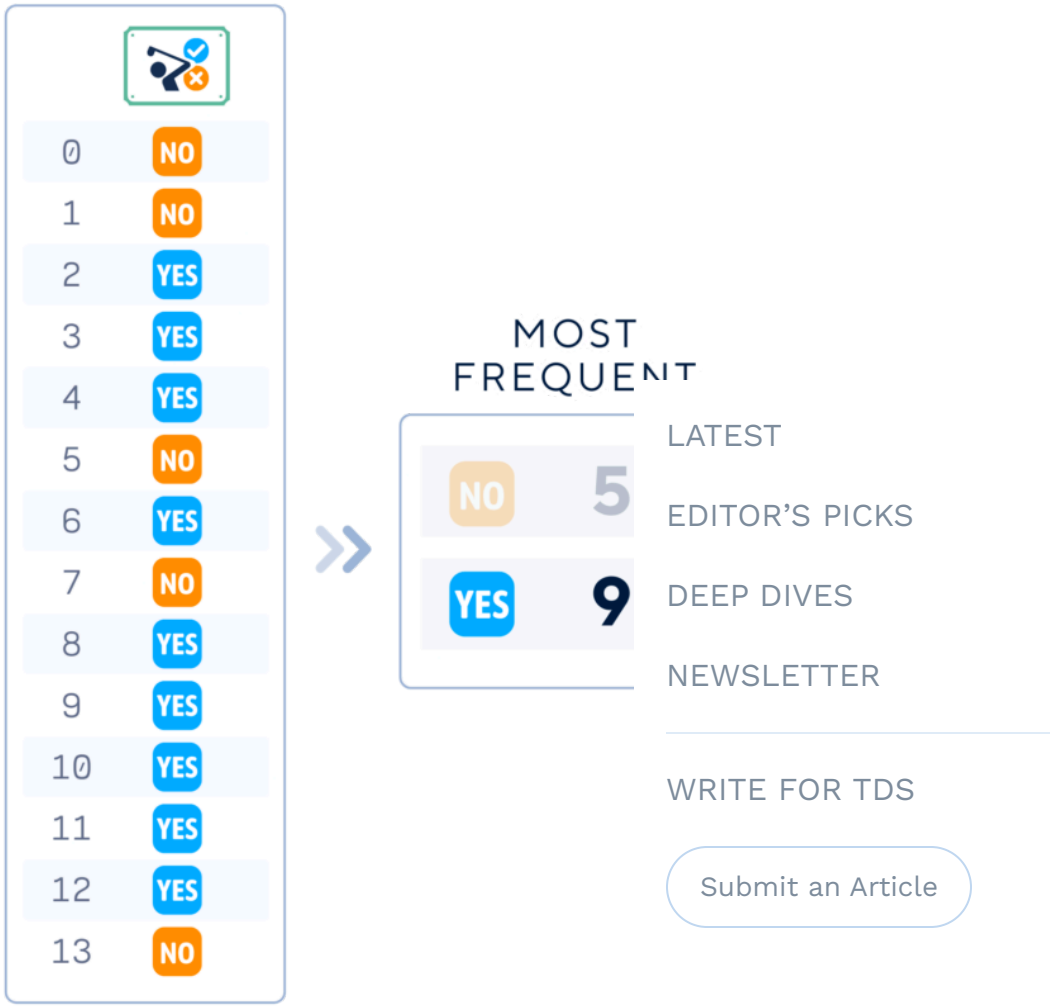


Depends on the strategy, Dummy Classifier makes different


```
from sklearn.dummy import DummyClassifier

# Choose a strategy for your DummyClassifier (e.g.,
strategy = 'most_frequent'
```

2. Collect Training Labels

Collect the class labels from the training datase
the strategy parameters.

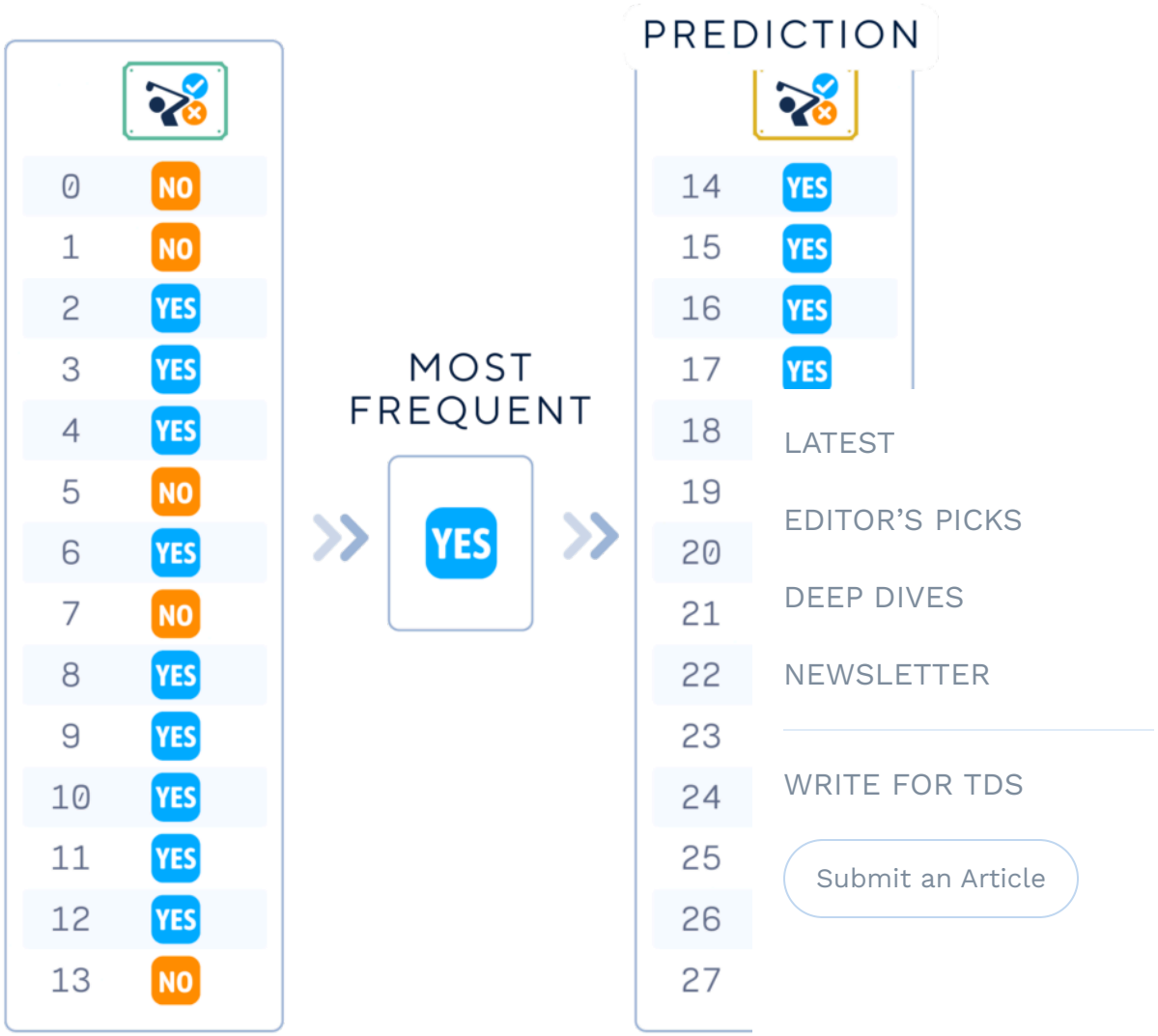


The algorithm is simply getting information of the "Most Frequent" class in the "Yes".

```
# Initialize the DummyClassifier
dummy_clf = DummyClassifier(strategy=strategy)

# "Train" the DummyClassifier (although no real trai
dummy_clf.fit(X_train, y_train)
```

3. Apply Strategy to Test Data

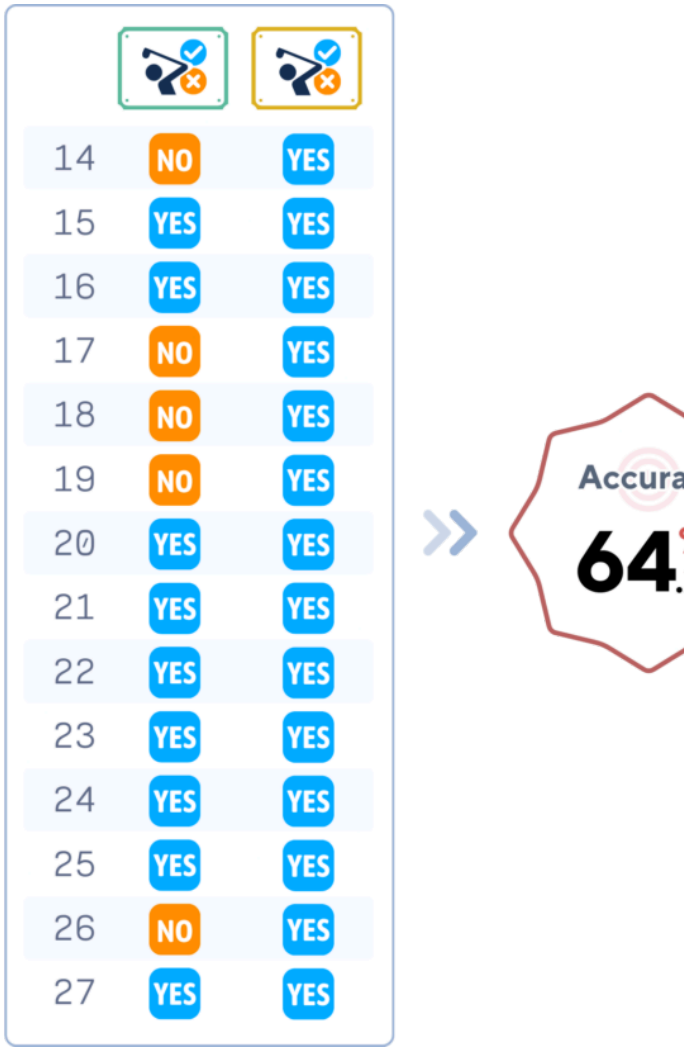
Use the chosen strategy to **generate a list of pre**
your test data.



If we choose the "most frequent" strategy and find that "Yes" (play golf) appea
data, the dummy classifier will simply remember to always p

```
# Use the DummyClassifier to make predictions
y_pred = dummy_clf.predict(X_test)
print("Label      :",list(y_test))
print("Prediction:",list(y_pred))
```

Evaluate the Model



Dummy classifier gives 64% accuracy as the baseline for futu


```
# Evaluate the DummyClassifier's accuracy
from sklearn.metrics import accuracy_score

accuracy = accuracy_score(y_test, y_pred)
print(f"Dummy Classifier Accuracy: {accuracy.round(4
```

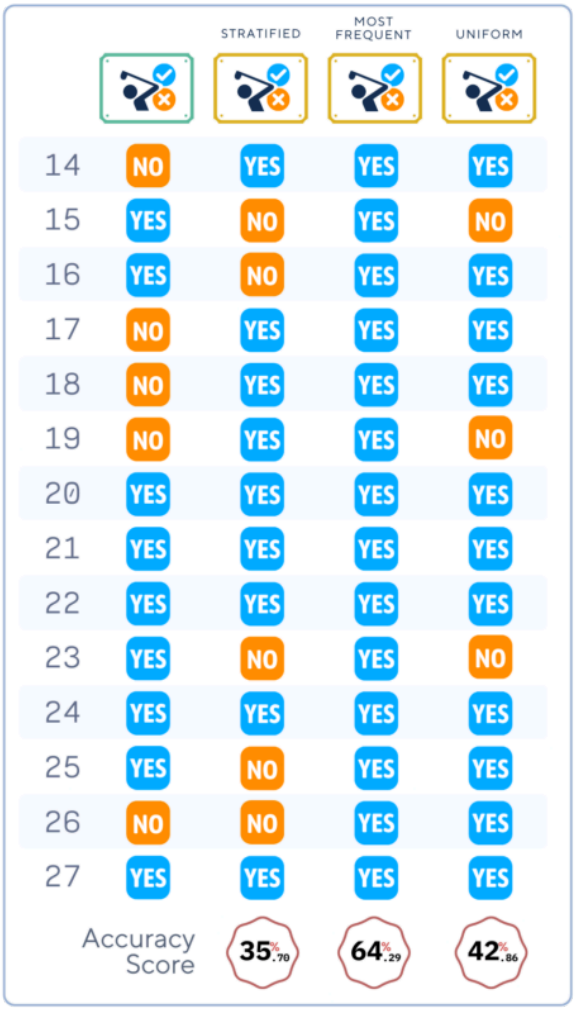
Key Parameters

While dummy classifiers are simple, they do hav
important parameters:

1. **Strategy**: This determines how the classifier
   predictions. Common options include:

- ‘most_frequent’: Always predicts the most common class in the training set.
- ‘stratified’: Generates predictions based on the training set’s class distribution.
- ‘uniform’: Generates predictions uniformly at random.
- ‘constant’: Always predicts a specified class

2. **Random State**: If using a strategy that involv
(like ‘stratified’ or ‘uniform’), this parameter
reproducibility of results.
3. **Constant**: When using the ‘constant’ strateg
specifies which class to always predict.



For our golf dataset, we might choose the ‘most_frequent’ strategy, which
parameters.

Pros and Cons

Like any tool in machine learning, dummy classifiers have their strengths and limitations.

Pros:

1. **Simplicity**: Easy to understand and impleme
2. **Baseline Performance**: Provides a minimum benchmark for other models.
3. **Overfitting Check**: Helps identify when com overfitting by comparing their performance classifier.
4. **Quick to Train and Predict**: Requires minima resources.

Cons:

1. **Limited Predictive Power**: By design, it does data, so its predictions are often inaccurate
2. **No Feature Importance**: It doesn't provide ir features are most important for predictions
3. **Not Suitable for Complex Problems**: In real- with intricate patterns, dummy classifiers a to be useful on their own.

Final Remarks

Understanding dummy classifiers is crucial for a or machine learning enthusiast. They serve as a

helping us ensure that our more complex models are actually learning useful patterns from the data. As you continue your journey in machine learning, always remember to compare your models against these simple baselines – you might be surprised by what you learn!

🌟 Dummy Classifier Code Summari

```
# Import necessary libraries
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.dummy import DummyClassifier

# Make dataset
dataset_dict = {
    'Outlook': ['sunny', 'sunny', 'overcast', 'rain'
    'Temperature': [85.0, 80.0, 83.0, 70.0, 68.0, 65
    'Humidity': [85.0, 90.0, 78.0, 96.0, 80.0, 70.0,
    'Wind': [False, True, False, False, False, True,
    'Play': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No',
}
df = pd.DataFrame(dataset_dict)

# Perform one-hot encoding on 'Outlook' column
df = pd.get_dummies(df, columns=['Outlook'], prefix=

# Convert 'Wind' and 'Play' columns to binary indica
df['Wind'] = df['Wind'].astype(int)
df['Play'] = (df['Play'] == 'Yes').astype(int)

# Split data into features (X) and target (y), then
X, y = df.drop(columns='Play'), df['Play']
X_train, X_test, y_train, y_test = train_test_split(

# Initialize and train the dummy classifier model
dummy_clf = DummyClassifier(strategy='most_frequent'
```

```
dummy_clf.fit(X_train, y_train)

# Make predictions on the test data
y_pred = dummy_clf.predict(X_test)

# Calculate and print the model's accuracy on the test data
print(f"Accuracy: {accuracy_score(y_test, y_pred)*100:.4f}%")
```


Further Reading

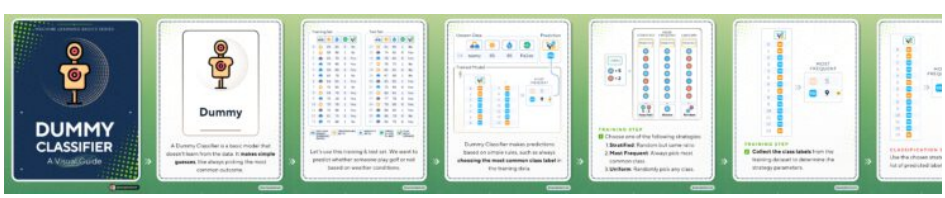
For a detailed explanation of the DummyClassifi
implementation in scikit-learn, readers can refe
documentation [2], which provides comprehensi
its usage and parameters.

Technical Environment

This article uses Python 3.7 and scikit-learn 1.5.
concepts discussed are generally applicable, sp
implementations may vary slightly with different

About the Illustrations

Unless otherwise noted, all images are created
incorporating licensed design elements from Ca



For a concise visual summary of Dummy Classifier, check out the comp

Reference

[1] T. M. Mitchell, Machine Learning (1997), McGraw-Hill Science/Engineering/Math, pp. 59

[2] F. Pedregosa et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://scikit-learn.org/stable/modules/generated/sklearn.dunfier.html

Classification Algorithms

WRITTEN BY

Samy Baladram

See all from Samy Baladram

Benchmarking | Classification Algorithms | Machi

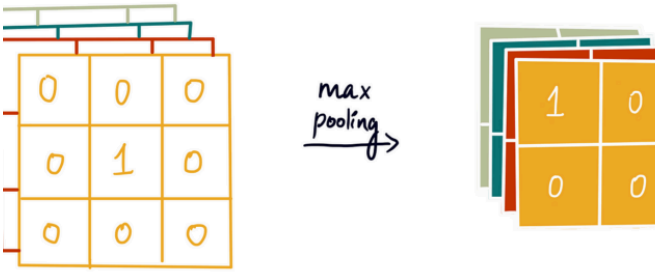
Programming | Visual Guide

Share This Article

Towards Data Science is a community publication. Submit your insights to reach our global audience and earn through the TDS Author Payment Program.

Write for TDS

Related Articles



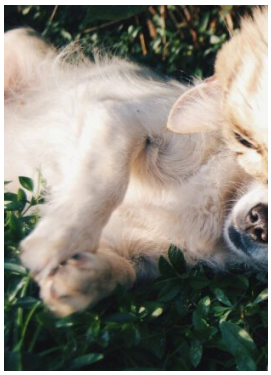
ARTIFICIAL INTELLIGENCE

Implementing Convolutional Neural Networks in TensorFlow

Step-by-step code guide to building a Convolutional Neural Network

Shreya Rao

August 20, 2024 6 min read



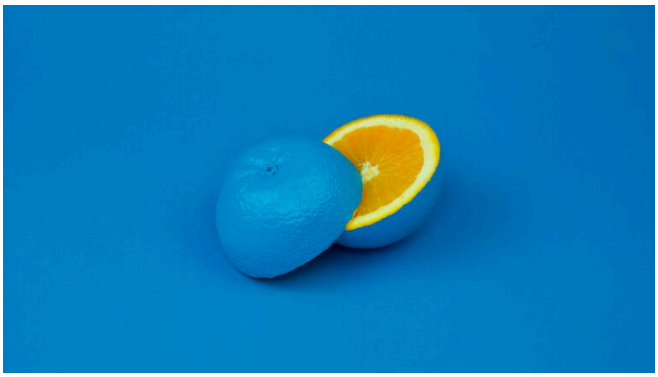
ARTIFICIAL INTELLI

How to Forec Time Series

A beginner's guid reconciliation

Dr. Robert Kübler

August 20, 2024



DATA SCIENCE

Hands-on Time Series Anomaly Detection using Autoencoders, with Python

Here's how to use Autoencoders to detect signals with anomalies in a few lines of...

Piero Paialunga

August 21, 2024 12 min read

MACHINE LEARNING

3 AI Use Cases (That Are Not a Chatbot)

Feature engineering, structuring unstructured data, and lead scoring

Shaw Talebi

August 21, 2024 7 min read

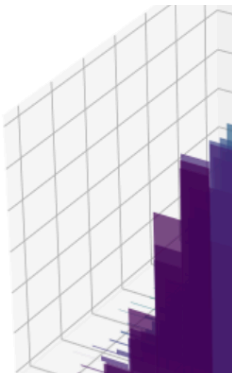
DATA SCIENCE

Back To Basics, Part Uno: Linear Regression and Cost Function

An illustrated guide on essential machine learning concepts

Shreya Rao

February 3, 2023 6 min read



DATA SCIENCE

Must-Know in Bivariate Norm Explained

Derivation and pr powerful concep

Luigi Battistoni

August 14, 2024 7

DATA SCIENCE

Our Columns

Columns on TDS are carefully curated collections of posts on a particular idea

or category...

TDS Editors

November 14, 2020 4 min read

LATEST

EDITOR'S PICKS

DEEP DIVES

NEWSLETTER

WRITE FOR TDS

Submit an Article

Your home for data science and AI. The world's leading publica
data analytics, data engineering, machine learning, and artificia
professionals.

© Insight Media Group, LLC 2025

Subscribe to Our Newsletter

WRITE FOR TDS · ABOUT · ADVERTISE · PRIVACY POLIC

COOKIES SETTINGS